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LETTER

# Large language models in climate and sustainability policy: limits and opportunities

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## Abstract

Accurate, reliable and updated information support effective decision-making by reducing uncertainty and enabling informed choices. Multiple crises threaten the sustainability of our societies and pose at risk the planetary boundaries, hence requiring usable and operational knowledge. Natural-language processing tools facilitate data collection, extraction and analysis processes. They expand knowledge utilization capabilities by improving access to reliable sources in shorter time. They also identify patterns of similarities and contrasts across diverse contexts. We apply general and domain-specific large language models (LLMs) to two case studies and we document appropriate uses and shortcomings of these tools for two tasks: classification and sentiment analysis of climate and sustainability documents. We study both statistical and prompt-based methods. In the first case study, we use LLMs to assess whether climate pledges trigger cascade effects in other sustainability dimensions. In the second use case, we use LLMs to identify interactions between the sustainable development goals and detects the direction of their links to frame meaningful policy implications. We find that LLMs are successful at processing, classifying and summarizing heterogeneous text-based data helping practitioners and researchers accessing. LLMs detect strong concerns from emerging economies in addressing food security, water security and urban challenges as primary issues. Developed economies, instead, focus their pledges on the energy transition and climate finance. We also detect and document four main limits along the knowledge production chain: interpretability, external validity, replicability and usability. These risks threaten the usability of findings and can lead to failures in the decision-making process. We recommend risk mitigation strategies to improve transparency and literacy on artificial intelligence (AI) methods applied to complex policy problems. Our work presents a critical but empirically grounded application of LLMs to climate and sustainability questions and suggests avenues to further expand controlled and risk-aware AI-powered computational social sciences.

## 1. Introduction

The intertwined nature of multiple global crises is threatening the stability of our planet and is posing challenges to our societies. Direct and cascading climate impacts are provoking biodiversity collapse and ecosystem destruction, disinformation is posing democracy under attack and military and economic wars are redesigning the geopolitical landscape [1]. In the face of these issues, digital technologies, if properly designed and used, could support sustainability transitions by increasing the efficiency of the production and consumption modes [2], democratizing knowledge through access to information [3] and by launching new scientific breakthroughs [4]. The internet of things monitor in quasi real-time the state of our connected world, while machine learning and artificial intelligence (AI) process and elaborate

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the wealth of information produced to generate new insights and knowledge.

In the realm of sustainability, environmental, economic, and social dimensions exerts concurrent influences on both anthropogenic and natural ecosystems. The interactions across different domains can be positive or negative, leading to synergies and trade-offs across different time and spatial scales. The study of these linkages has recently gained popularity: researchers have engaged in global [5], regional [6] and sectorial [7] analyses to turn interactions into actions. These efforts are possible also thanks to the 17 sustainable development goals (SDGs), launched by the United Nations (UN) in 2015 and operationalized through 149 targets and 281 quantitative indicators. The SDGs offer a measurable and traceable universal roadmap for the near future (up to 2030) and detail what is at the heart of sustainable development. Since their establishment, the SDGs have spurred funding, legislation initiatives and policy prescriptions worldwide generating text-based inputs heterogeneous per nature, scope and aim. The analysis of these documents reveals different degrees of ambitions [8], critical gaps [9] and political priorities [10]. However, the elicitation of policy-relevant knowledge is often left to expert groups, extensive manual discourse and content analyses with hard-to-replicate routines [11]. Oftentimes, findings are hard to elucidate as complex trade-offs and synergies are involved. To overcome selection biases and to boost open science practices, the use of AI-based approaches—mainly natural-language processing (NLP)—has become prominent [12]. The widespread penetration of large language models (LLMs) supports this trend and opens the door to hybrid methods which integrate human and machine-based insights [13, 14]. As more studies flourish in this space, so do critiques [15, 16]. The strongest fronts call for more transparent, open and explainable models [17, 18], but there are growing concerns about AI's environmental footprint and costs [19, 20]. Beyond the sustainability and ethical considerations of the use of LLMs (and generative AI in general), the real-life usability of AI outputs for public policy and policy agenda setting remains loosely addressed by scholars. This knowledge gap largely stems from the ongoing transformation AI affects within the public policy sphere. Empirical data about adoption rates, usage frequency and real-world deployment remain constrained by limited spatial and temporal resolution. Nevertheless, as AI continues to attract attention from researchers and practitioners, the formulation of robust frameworks for integrating AI-generated insights into public policy emerges as an urgent imperative to mitigate the risk of misuse [21]. When compounded with other socio-environmental crises—such as the inadequate responses to climate change and the shortfall in achieving the SDGs—an unregulated and a risk-insensitive application of AI to public policy may trigger cascading adverse effects across interconnected systems. In this work, we provide an empirical and critical overview of the use of LLMs for climate and sustainability policy purposes by investigating the process of two widely used tasks: text classification and sentiment analysis. We document opportunities, challenges and assumptions in applying NLP and LLMs to two large-scale analyses which use different text-based corpora. Both case studies aim at supporting policy-makers with actionable insights to maximize synergies and minimize trade-offs between different sustainability dimensions. This work responds to a growing call for transparent [22, 23] and trustworthy [24, 25] AI-augmented knowledge production chain and it complements ongoing assessment efforts to discuss specific issues associated with AI in public policy [26].

## 2. Methods

### 2.1. The context: alignment as a key policy objective

We present the methodological framework underlying two parallel case studies conducted between September 2023 and September 2024. Both case studies leverage heterogeneous textual data as input and aim to advance the state of the art in examining synergies and trade-offs among potentially conflicting policy objectives, improving the alignment between them.

In the first case study, we focus on climate pledges as key inputs to determine the type and extent of alignment between climate action and other SDGs. In its Article 4, the 2015 Paris Agreement requires that each party prepares, explains and shares how it intends 'to limit global average temperature increases to as close to 1.5 °C as possible'. The NDCs meet that request: they are political documents, submitted and updated every five years, containing pledges to pursue climate mitigation and adaptation actions. Individual submissions define the degree of ambition of each party. As a collective corpus, instead, the NDCs determine whether the world is on track to achieve the long-term goals of the Paris Agreement. As pathways to change are by definition developed overtime, the analysis of submissions becomes a powerful tool to assess diverging or converging behaviors with respect to climate policy. The NDCs have received considerable attention as a coordination mechanism [27] and governance instruments [28]. As they explore how different sectors, groups and areas of an individual party will have to adjust to meet climate ambitions, the NDCs also embed other important political priorities which affect the sustainable development agenda The term 'sustainable development' appeared for the first time in the famous Brundtland report in 1987 [29]. In there, 'sustainable development' is defined as the ability of humanity to ensure that the needs of the present do not compromise 'the ability of future generations to meet their own.' While being a concrete

aspiration, 'sustainable development' is also a multi-faceted and complex concept.

Since the publication of the Brundtland report, it lays on four critical dimensions: promoting economic growth, ensuring the respect and protection of the environment to avoid degradation, enabling everyone to fulfil her/his potential within communities, and supporting adequate institutions [29]. Policies (meant as the set of actions decision-takers put in place) supporting sustainable development may cover individual dimensions or them all. They can change over time and—depending on specific contextual factors—they can target certain groups more than others. This high degree of heterogeneity is also a challenge to policy makers, which soon after the launch of the Brundtland report realized the world needed an operational guidance to fulfill the promises of sustainable development. Launched in Addis Ababa 2030 Agenda for Sustainable Development, the SDGs cover 17 goals and they represent a global framework for the near-term future. Currently, the SDGs constitute the backbone of action towards a sustainable 'common future': they shape the channels through which plans are designed and decisions are taken. As they describe pathways of change through their goals, targets and indicators, the SDGs are the most suitable tools to represent how different sustainability dimensions interact with one another. Their interlinkages (positive or negative) describe this complex landscape and—for this reason—they are the preferred lens through which the sustainable development agenda takes shape [30, 31]. When climate and sustainability objectives are misaligned, the resulting gap creates significant hurdles that slow or potentially derail mitigation and adaptation efforts [32], while also adversely affecting other socio-economic and environmental goals [33, 34].

However, a misalignment can also exist among different SDGs themselves. Therefore, identifying synergies (i.e., mutually reinforcing strategies) and trade-offs (i.e., actions that are sub-optimal from a Pareto-efficiency standpoint) across the multidimensional sustainability landscape constitutes a policy challenge in its own right. To support policymakers with actionable insights about alignments between conflicting goals, we run the second study applying LLMs to a corpus of heterogeneous documents. Our goal is to detect the critically neglected areas of action (i.e. isolated goals) and to recognize patterns across multiple SDGs by assessing the strength of recurring connections. Furthermore, we aim at detecting the direction of the links to suggest how to prioritize interventions and redirect funds and efforts when needed.

### 2.2. Data collection

The study of synergies and trade-offs across sustainability domains has gained traction through the assessment of published evidence on SDG interlinkages. Given the heterogeneity in terms, no query will ever cover the entire universe. The literature has approached this problem by involving experts to pre-select studies based on their knowledge [5]. Over the past two decades, this trend has raised questions about the risk of oversimplifying reality [35],the need for credible, salient and legitimate information [36] and the representativeness of the knowledge systems these experts cover [37]. Calls to broaden inclusivity by using collaborative platforms, automated routines [38] and AI [39] have appeared, but the institutional processes are still digesting these routines and are far from implementing them. Instead, the dominant view lies on the creation of Expert Groups (e.g., the UNDESA expert groups meetings for every SDG), which regularly meet across different sustainability dimensions or focused solely on specific areas.

For the first case study, we focus on a specific universe: climate pledges downloaded in PDF format from the official UNFCCC NDC registry [40]. We focus on the first two submission rounds of the NDCs to zoom into the content of declared pledges of change. The focus on political documents has a forward-looking purpose: we move beyond assessing positive or negative interlinkages between the climate and development agenda and we present what the near-term future might look like if pledges were fully met. By doing so, we proactively signal where critical misconnections are and we inform the new wave of NDC submissions. For the second case study, we start with the universe definition to overcome the limits of potentially partial queries. We use a collaborative approach named Delphi method (see supplementary materials) to determine the most relevant types of documents using an iterative process. The Delphi Method led to the identification of key document types, which span civil society, politics and communication. Through three rounds of the Delphi method, we identify the following documents of interest:

i) National reports on SDG progress: these submissions—named voluntary national reviews—are meant to demonstrate progress in achieving the SDGs. They are non-mandatory and non-binding, but they rather provide a platform to showcase where efforts were allocated. In our framework, these documents represent political actions. We include 328 VCRs;
ii) ii) civil society initiatives around the SDGs: we include 8015 actions as voluntary submitted by a broad range of actors (i.e., governments, NGOs, UN agencies, etc.) on the SDG Actions platform [41];
iii) news items as downloaded from the SDG Knowledge Hub [42], capture the pulse around sustainability across countries and SDG actions. We include 8865 items.

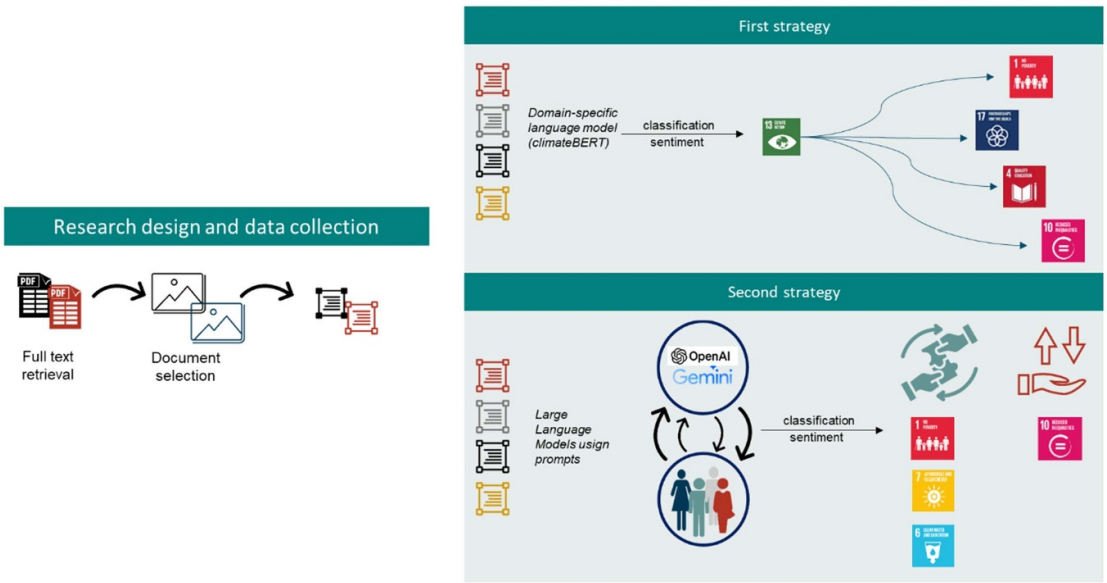


**Figure 1.** For both case studies we test two strategies to accomplish the classification and sentiment tasks. The first entails domain-specific language models and the use of statistical methods (i.e. semantic similarity). The second involved the deployment of LLMs and prompts. Prompts are designed using a trial-and-error approach and refined over several turns.

### 2.3. The AI routine

For both studies and prior to text processing, we developed an AI-based routine testing two parallel workflows (figure 1) involving classification and sentiment as main tasks. The first strategy deploys domain-specific language models for bothclassification and sentiment analysis using ClimateBERT, a climate-domain-adapted LLM [13]. ClimateBERT uses DistilROBERTA as a starting point and it builds on additional training using corporate disclosures, climate-related paper abstracts and private companies' reports. Downstream models include—among others—the classification of climate-related sentiments in climate-focused paragraphs according to three categories: 0 if the paragraph contains negative text with respect to climate adaptation and mitigation; 1 if it is neutral and 2 if there is an explicitly positive recall to actions which may be positive to climate adaptation and mitigation. We were interested in detecting paragraphs tackling one or more specific SDGs and to derive whether these were expressing positive or negative viewpoints on climate adaptation and mitigation. The second strategy entails the deployment of general LLMs, in our specific case Gemini 1.5 Flash by Google Research [43], and the design of a prompt-based routine to achieve the research objectives. Gemini 1.5 Flash was used through the Google AI Studio application program interface and the library *generativeai*.

The first strategy determined the length of the textual unit of interest: as climateBERT-sentiment is trained on paragraphs, we splitted the full-length NDCs into meaningful paragraph-long chunks (see supplementary materials). Once we obtained the clean and workable text, we proceeded to the SDG detection for every paragraph. We computed semantic similarity between every paragraph and the SDGs as derived from the UN SDG Fast Facts [44]. We used this description instead of any other alternative because it contains a more policy-friendly style and departs from pure academic language. We assigned to each paragraph the SDG with maximum similarity score and we then passed the so-obtained paragraphs into ClimateBERT-sentiment (figure 2).

For our second strategy, we developed three versions of the same prompt, and we applied it to two performance-equivalent models (GPT3.5 and Gemini 1.5 Flash). Each version carries a slight but meaningful-relevant modification (table 1). Initially, we asked the models to reveal the top three SDGs, only focusing on three diverse classes: quantity (*dominance*), value (*relevance*), and a mix of the two (*prominence*). Dominance refers to the appearance of concepts and words in a given text. We consider it as more context-agnostic in this setting. Relevance is context-relevant and includes value judgments and value assessments of individual concepts. Even when less abundant in the text, some words can leave a mark on the reader because they are at the heart of a whole message. Prominence, in this contest, is a mixed area where words and concepts are both heavily cited and relevant.

We run each prompt three times on a sample of paragraphs as derived by the NDCs in random order. Our aim was twofold. First, we wanted to check whether the models capture the marginal but meaning-relevant modifications providing different results. Second, we aimed to assess the robustness of the models when identifying the relevant SDGs

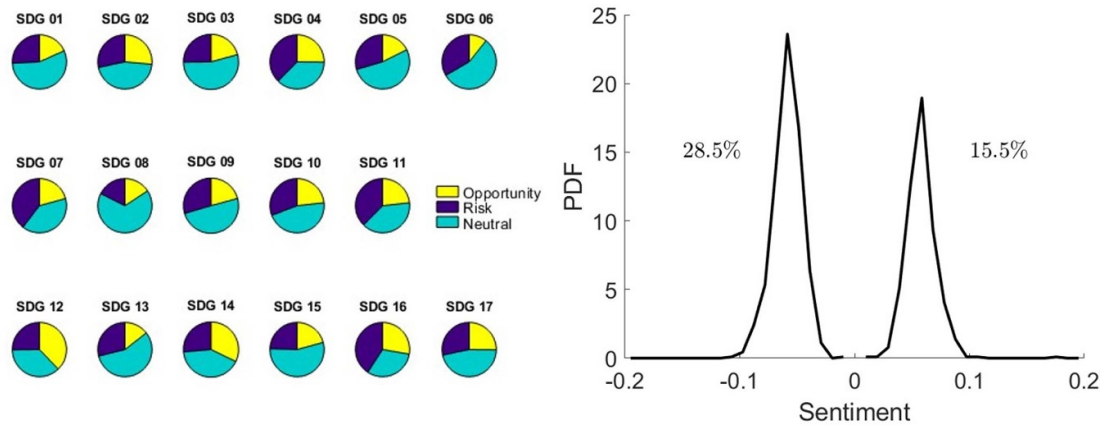


**Figure 2.** Left panel: The sentiment and SDG allocation of the first strategy. SDGs are detected using semantic similarity between NDCs' paragraphs and SDG Fast Facts downloaded from the UN website. Right panel: probability density function (PDF) of the expected value of sentiment, obtained by combining sentiment (between −1 and 1, risk and opportunity) and allocation score (0–1). Neutral cases are removed, as their sentiment is 0. In this case, SDG13, even if the maximum of both sides of the PDF are similar ([sentiment] = 0.35), the larger area in the negative region indicates risk.

**Table 1.** Alternative prompts.

| | |
|---|---|
| Prompt #1: quantity-oriented | 'Assign the following text to the top three SDGs based on their **dominance**' |
| Prompt #2: value-oriented | 'Assign the following text to the top three SDGs per **relevance**' |
| Prompt #3: quali-quantity oriented | 'Assign the following text to the top three SDGs based on their **prominence**' |

irrespective of the order of paragraphs. We find differences between runs (for both models) and prompt versions (solely in the GPT3.5 case). While differences between prompt versions are expected—as questions differ and capture different aspects of the same problem—the order of appearance across paragraphs should not affect the classification of each text to important SDGs. As Gemini 1.0 maintains consistency across all runs and is not subject to randomness, we decided on Gemini 1.0 as the appropriate model for the study. We built a two-step prompt structure (see supplementary materials, table 1) to capture relevant SDGs per paragraph and the tone of these connections. We avoid fine-tuning and domain-specific training as Gemini already includes knowledge about the SDGs. The prompt-engineering process benefitted from a trial-error ping-pong exchange between the research team members. The final prompt was reached after eight attempts. We find different sentiments across countries, with respect to social, economic of environmental SDGs (figure 3). As an automatic evaluation is not feasible because of the lack of benchmark datasets in this field, we engaged in a manual screening of both SDG allocations and sentiments.

## 3. Limits and opportunities of the framework

Some of the biggest challenges associated with the present framework lie in the data collection and labeling (figure 4, Step 1). First, peer-review works take time to publish and largely focus on identifying alignments across different policy agendas finding mainly synergies (especially within SDGs). This skewed distribution of the results is motivated by the genuine desire to support policy with proactive recipes for change rather than exposing bottlenecks. Furthermore, the literature on local evidence is scattered and hard to access: grey literature and ad-hoc reports are dominated by context-specific findings which may not always find space in international journals. As there is no global database for such documents yet, individual researchers must make choices which profoundly affect the outcome of their scientific enquiry. A large body of evidence can be found in projects' deliverables, financed activities and even press releases. These insights shall not be neglected as they capture the reality on the ground. Second, documents may often lack accessibility due to restrictive paywalls. This is especially relevant for disadvantaged institutions and contexts where large-scale agreements do not exist. Open-science practices are not the standard yet and the least developed countries suffer greatest limits. The same issues apply to corporate reports, which are offered under subscription and license agreements. Depending on data and text cleaning and retrieval routines, the outcome may be biased threatening external validity, meant as the extent to which results may be adequately generalized. Limits to interpretability are also posed by the queries under use. This is relevant for linguistic purposes and for issues which change overtime. An example is climate policy research, which has evolved over the

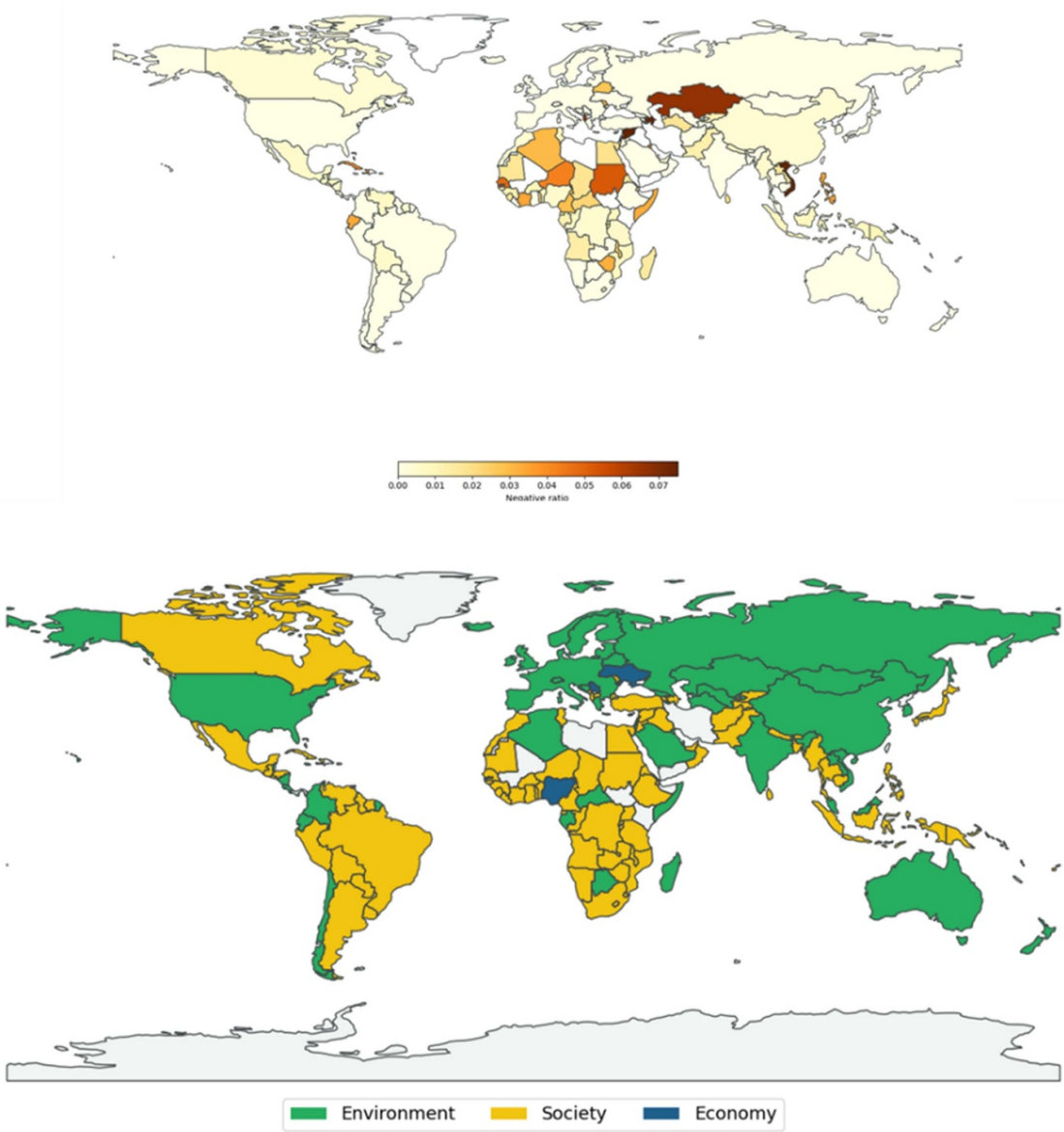


**Figure 3.** We run a prompt-based sentiment analysis to assess the nature of the NDC-SDG connections. We then compute the dominant sentiment per country using a Negative-to-Positive ratio (upper). Emerging and developing economies are less optimistic than countries in the Global North in presenting the challenges and opportunities of the green transition and climate action. Bottom-part: we group SDGs according to their global area of interest mirroring previous literature in the field [2, 45]. We find that environmental issues (biodiversity protection, forest management and water bodies) are primary issues for developed economies, while emerging and developing countries flag food security, water and sanitation and the economics of the energy transition as urgent matters. On the environment category: SDG6, SDG13, SDG14 and SDG15; in the society category: SDG1, SDG2, SDG3, SDG4, SDG5, SDG7, SDG11, SDG16; within the economy category: SDG8, SDG9, SDG10 and SDG12. We included SDG17 in the social category.

past four decades. Policies were centered on 'global warming' and 'greenhouse gas effect' back in early 2000s. As the science has evolved, so has the way we refer to these phenomena, favoring 'climate change', 'climate collapse' and 'climate crisis'. A comprehensive query aiming at capturing the full spectrum of a given sustainability domain may need to adopt all three modifications.

Beyond data collection, data processing (figure 4, Step 2) poses the second challenge in applying automated routines (and particularly LLMs) to sustainability and climate policy documents. In the presence of big literature and big data queries, researchers may need to classify documents (according to temporal and spatial dimensions) removing those documents which do not seem fit and may skew distributions of topics, issues or themes of interest. This data collection procedure is different from text cleaning and text retrieval processes. Meaning can change depending on the unit under consideration: the full document conveys a message which may be different from the one presented in a single sentence, paragraph or chapter. In the process of analysis of SDG interlinkages, these differences become crucial

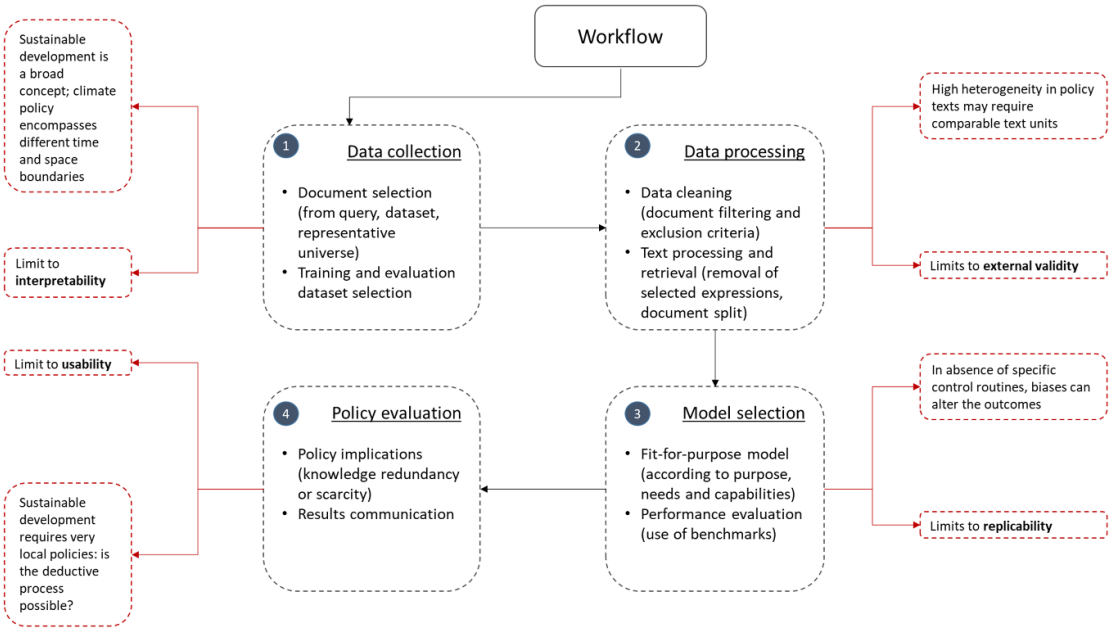


**Figure 4.** The application of LLMs to sustainability and climate policy documents in a four-step workflow. Each step presents a specific threat (or limit).

and require strong reflections about the optimal context-window length. Text cleaning from special characters, recurring expressions, numbers and selected linguistic forms constitutes an important step in NLP applications (mostly frequency-based statistical approaches). LLMs (i.e., from GPT3 onwards) are less affected by irrelevant expressions as the self-attention mechanism enables modeling the influence of every word on the others in parallel, through the computation of an attention score what every word means in context aided by a wider and more elaborate context-window. Still, data cleaning remains a good practice to enable full-scale reproducibility and involves also taking decisions about inclusion and exclusion rules of figures and tables headings and captions, titles and subtitles and focus boxes.

Both interpretability and external validity are limits encountered and documented in the use of LLMs in clinical and medical research [46, 47]. In the sustainability and climate policy domains, we acknowledge two additional risks identifiable in the model selection and output evaluation stages (figure 4, Step 3). The choice of the most suitable and fit-for-purpose model is conditioned by market forces and four other factors. There are plenty of LLMs available and released at impressive speed [48]. Literature has classified LLMs according to the their neural architecture [48], pre-training method [49], accessibility, general purpose and domain-specific type, multilingual and mono, bilingual abilities [50], among others. The choice of the model used to solve the specific research question depends on several factors:

- Technical—context window, prompting technique and available computing resources.
- Economic—open-source and open-access models are preferred alternatives, especially in budget-constrained contexts.
- Policy—interpretable outputs and full-scale replicability are core needs and the use of 'black boxes' needs to be avoided [17].
- Purpose: the objective is highly context-or domain-specific and requires fine tuning.

As models are published and used, evaluations and applications flourish. Platforms such as HuggingFace OpenLLM Leaderboard provide benchmark evaluations to assess model capabilities in a standardized way, but specialized knowledge is required to choose over the extensive number of model. When certain models exhibit worse performance than others, they may be dismissed. Finally, certain models may be withdrawn because of issues encountered in their initial deployments or simply retired and not made available anymore (e.g. Gemini 1.0). These market-driven factors, together with the technical, economic, policy and purpose motives, put at risk the reproducibility of results and ultimately the legacy of the derived policy implications. Prompting techniques in sustainability and climate policy analyses overcome these challenges in two ways. Prompts support replicability and they can be designed to respond to specific purposes. Even in absence of a specific model, the process and workflow is documented. As prompts are typically in use for general-purpose LLMs—trained on billion parameter scale large text corpora—their use surveys the world wide available knowledge. To

reach their objective, prompts must be specific, calibrated on the research question and elaborated enough to cover the objective. Prompts in sustainability and climate policy may need refinement over several iterations and they may involve multiple experts in domain-specific knowledge compartments.

The final stage of the workflow relates to the translation from useful to usable science, from research into policy and action (figure 4, Step 4). This step is probably the most complex as it involves different skillsets. Communication of the results is the first milestone: as there is a gap between what researchers understand as useful and what users recognize as such [51], the presentation of the insights gathered from the research shall be co-produced with relevant stakeholders. When appropriate, uncertainties in the model outputs and human-made decisions which potentially lead to risks and threats shall be made explicit. Limits to usability of the outputs are also imposed by the scale of the outputs. Global or regional findings can provide information on trends and patterns of change. However, local decision-makers may require higher granularity to design policies which catch the specificities of the context. The effort to link global, regional and local settings must come from the researchers themselves and represents an important portion of the task.

We find that there is a need for urgent research and development in three core areas. First, the use of LLMs in sustainability and climate policy cannot be separated from the development of transparent output-evaluation methods. In tasks such as text classification, sentence similarity and zero-shot classifications the assessment of LLM results is an important step to assess whether some dimensions or event topics are extensively or poorly tackled by certain policy interventions. The abundance of specific areas in policy documents can—for example—lead to stronger funding allocations as policy makers can perceive them as urgent and prioritize those issues. On the other end, topics which appear less urgent can remain critically left out, exacerbating inequalities and unjust dynamics. Second, LLMs extract meaningful insights considering the semantic and syntactic processes which give birth and shape to a text. LLMs are also trained on large-scale corpora. However, the assumption that LLMs grasp the meaning of policy documents is only partially correct. Being the outcome of compromise, negotiations and often long iterative processes, policy texts are expressions of what cannot be made explicit: intent and hidden motives. In the climate domain this anecdotal evidence has been widely documented [52] and raises questions about the actual comparability between documents produced in different moments or signed and put forward by different countries, departments or agents [53]. Third, the use of LLMs requires significant human interventions and decisions which may often be overlooked. While the use of keyword-based methods has been superseded and conceived as strongly biased, the deployment of prompts may suffer from the very same threats. In one of our applications, we tested how subtle but meaningful differences play a role in deciding over the use of one LLM over another. These details are far from being insignificant in policy processes: they make their mark in interpretation of the results. Equally important, we noted that while the launch of LLMs may require limited specialized knowledge, the sense-making of the insights they produce calls for highly expert-based skills. Therefore, we advocate for a new paradigm where human-aware machine-enabled insights become an integral part of the policy processes.

## Data availability statement

All data that support the findings of this study are included within the article (and any supplementary files).

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## Conflict of interest

The authors declare no competing interests, financial or other, exist.

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## References

[1] Søgaard Jørgensen P *et al* 2024 Evolution of the polycrisis: anthropocene traps that challenge global sustainability *Phil. Trans. R. Soc. B* **379** 20220261
[2] Vinuesa R, Azizpour H, Leite I, Balaam M, Dignum V, Domisch S, Felländer A, Langhans S D, Tegmark M and Fuso Nerini F 2020 The role of artificial intelligence in achieving the Sustainable Development Goals *Nat. Commun.* **11** 233
[3] Koldunov N and Jung T 2024 Local climate services for all, courtesy of large language models *Commun. Earth Environ.* **5** 13
[4] Wang H *et al* 2023 Scientific discovery in the age of artificial intelligence *Nature* **620** 47–60
[5] Bennich T, Persson Å, Beaussart R, Allen C and Malekpour S 2023 Recurring patterns of SDG interlinkages and how they can advance the 2030 Agenda *One Earth* **6** 1465–76
[6] Kuc-Czarnecka M, Markowicz I and Sompolska-Rzechuła A 2023 SDGs implementation, their synergies, and trade-offs in EU countries—Sensitivity analysis-based approach *Ecol. Indic.* **146** 109888
[7] Fuso Nerini F *et al* 2018 Mapping synergies and trade-offs between energy and the Sustainable Development Goals *Nat. Energy* **3** 10–15
[8] Lam S *et al* 2021 How are climate actions evaluated? A review of United Nations food security evaluations *Glob. Food Secur.* **28** 100509
[9] Tompkins E L, Vincent K, Nicholls R J and Suckall N 2018 Documenting the state of adaptation for the global stocktake of the Paris Agreement *Wiley Interdiscip. Rev. Clim. Change* **9** e545
[10] Klinsky S 2023 How needs-based assessments could advance equity in the global stocktake and beyond *Nat. Clim. Change* **13** 1007–9
[11] Alcamo J *et al* 2020 Analysing interactions among the sustainable development goals: findings and emerging issues from local and global studies *Sustain. Sci.* **15** 1561–72
[12] Tripathi S, Bachmann N, Brunner M, Rizk Z and Jodlbauer H 2024 Assessing the current landscape of AI and sustainability literature: identifying key trends, addressing gaps and challenges *J. Big Data* **11** 65
[13] Webersinke N, Kraus M, Bingler J A, Leippold M and Erlangen-nuremberg F A U 2022 ClimateBERT: A pretrained language model for climate-related text (arXiv:2110.12010)
[14] Vaghefi S A *et al* 2023 ChatClimate: grounding conversational AI in climate science *Commun. Earth Environ.* **4** 480
[15] Wickberg A and Gärdebo J 2023 Computation, data and AI in Anthropocene history *Hist. Technol.* **39** 328–46
[16] Messeri L and Crockett M J 2024 Artificial intelligence and illusions of understanding in scientific research *Nature* **627** 49–58
[17] Vinuesa R and Sirmacek B 2021 Interpretable deep-learning models to help achieve the sustainable development goals *Nat. Mach. Intell.* **3** 926
[18] Bender E M, Gebru T, McMillan-Major A and Shmitchell S 2021 On the dangers of stochastic parrots: can language models be too big? ? *Proc. 2021 ACM Conf. on Fairness, Accountability, and Transparency* (Association for Computing Machinery) pp 610–23
[19] Gibney E 2022 How to shrink AI's ballooning carbon footprint *Nature* **607** 648
[20] Gaffney O *et al* 2025 The Earth alignment principle for artificial intelligence *Nat. Sustain.* **8** 467–9
[21] Benoit K 2024 AI and Data Science for Public Policy *LSE Public Policy Rev.* **3**
[22] Bail C A 2024 Can Generative AI improve social science? *Proc. Natl Acad. Sci.* **121** e2314021121
[23] Grimes M, von Krogh G, Feuerriegel S, Rink F and Gruber M 2023 From scarcity to abundance: scholars and scholarship in an age of generative artificial intelligence *Acad. Manage. J.* **66** 1617–24
[24] Gusenbauer M 2023 Audit AI search tools now, before they skew research *Nature* **617** 439
[25] Grossmann I, Feinberg M, Parker D C, Christakis N A, Tetlock P E and Cunningham W A 2023 AI and the transformation of social science research *Science* **380** 1108–9
[26] OECD 2024 Governing with artificial intelligence: are governments ready? *OECD Artificial Intelligence Papers, No. 20* (OECD Publishing) (https://doi.org/10.1787/26324bc2-en)
[27] Da Zhu J 2022 Cooperative equilibrium of the China-US-EU climate game *Energy Strateg. Rev.* **39** 100797
[28] Leiter T 2024 Nationally determined contributions (NDCs) as a governance instrument—accounting for politics, negotiation progress, and related mechanisms under the Paris Agreement *Environ. Polit.* **33** 552–7
[29] World Commission on Environment and Development 1987 Our common future ('The Brundtland Report') *United Nations* pp 52–55
[30] Pradhan P 2023 A threefold approach to rescue the 2030 Agenda from failing *Natl Sci. Rev.* **10** nwad015
[31] Carmona R, Reed G, Thorsell S, Dorough D S, MacDonald J P, Rai T B and Sanago G A 2023 Analysing engagement with indigenous peoples in the intergovernmental panel on climate change's sixth assessment report *npj Clim. Action* **2** 29
[32] Savin I, King L C and van den Bergh J 2025 Analysing content of Paris climate pledges with computational linguistics *Nat. Sustain.* **8** 297–306
[33] Fuso Nerini F, Mazzucato M, Rockström J, van Asselt H, Hall J W, Matos S, Persson Å, Sovacool B, Vinuesa R and Sachs J 2024 Extending the sustainable development goals to 2050—a road map *Nature* **630** 555–8
[34] Nerini F, Sovacool B, Hughes N, Cozzi L, Cosgrave E, Howells M, Tavoni M, Tomei J, Zerriffi H and Milligan B 2019 Connecting climate action with other sustainable development goals *Nat. Sustain.* **2** 674–80
[35] Stirling A 2010 Keep it complex *Nature* **468** 1029–31
[36] Cash D W, Clark W C, Alcock F, Dickson N M, Eckley N, Guston D H, Jäger J and Mitchell R B 2003 Knowledge systems for sustainable development *Proc. Natl Acad. Sci.* **100** 8086–91
[37] Nature Climate Change Editor 2011 Evolving the IPCC *Nat. Clim. Change* **1** 227
[38] De-Gol A J, Le Quéré C, Smith A J P and Aubin Le Quéré M 2023 Broadening scientific engagement and inclusivity in IPCC reports through collaborative technology platforms *npj Clim. Action* **2** 49
[39] Larosa F, Hoyas S, García-Martínez J, Conejero J A, Fuso Nerini F and Vinuesa R 2023 Halting generative AI advancements may slow down progress in climate research *Nat. Clim. Change* **13** 497–9
[40] UNFCCC UNFCCC NDC registry (available at: https://unfccc.int/NDCREG)
[41] United Nations 2025 SDG actions platform (UNDESA) (available at: https://sdgs.un.org/partnerships/browse)
[42] Wulff D U, Meier D S and Mata R 2023 Using novel data and ensemble models to improve automated labeling of Sustainable Development Goals (arXiv:2301.11353)

[43] Gemini Team Google 2025 Gemini : a family of highly capable multimodal models (arXiv:2312.11805)
[44] United Nations 2025 SDG explainer fast facts (available at: www.un.org/sustainabledevelopment/sdg-fast-facts/)
[45] Rockström J and Sukhdev P 2016 Sustainable development goals: how food connects all the SDGs (available at: https://www.stockholmresilience.org/research/research-videos/2018-08-22-how-food-connects-all-the-sdgs.html)
[46] Reddy S 2023 Evaluating large language models for use in healthcare: a framework for translational value assessment *Inform. Med. Unlocked* **41** 101304
[47] Ullah E, Parwani A, Baig M M and Singh R 2024 Challenges and barriers of using large language models (LLM) such as ChatGPT for diagnostic medicine with a focus on digital pathology—a recent scoping review *Diagn. Pathol.* **19** 43
[48] Zhou C *et al* 2023 A comprehensive survey on pretrained foundation models: a history from BERT to ChatGPT (arXiv:2302.09419)
[49] Liu P *et al* 2023 Pre-train, prompt, and predict: a systematic survey of prompting methods in natural language processing *ACM Comput. Surv.* **55** 1–46
[50] OECD 2023 Ai language models technological, socio-economic and policy considerations *Oecd Digit. Econ. Pap.* **352** 1
[51] Crochemore L *et al* 2024 A framework for joint verification and evaluation of seasonal climate services across socioeconomic sectors *Bull. Am. Meteorol. Soc.* **105** E1218–E36
[52] Calliari E, Surminski S and Mysiak J 2019 The politics of (and behind) the UNFCCC's Loss and damage mechanism *Loss and Damage from Climate Change. Concepts, Methods and Policy Options* ed R-B L S T S S L-B J Mechler (Springer) pp 155–78
[53] Larosa F and Wickberg A 2024 Artificial Intelligence can help loss and damage only if it is inclusive and accessible *npj Clim. Action* **3** 59